

ENERGY

Energy is a measure of a body or system's ability to do work.

There are various types of mechanical energy which are applied to movement.

KINETIC ENERGY

This is the energy possessed by a body when it is moving.

It is given by:

$$E_k = \frac{1}{2}mv^2$$

where E_k = the kinetic energy (J)
 m = the mass of the object (kg)
 v = the velocity of the mass (ms^{-1})

If a force acts on a nett body, it does work on it. In doing so there is a change in the kinetic energy of the body.

i.e.

$$W = \Delta E_k = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$$

where $\frac{1}{2}mv^2$ = the final kinetic energy of the body.
 $\frac{1}{2}mu^2$ = the initial kinetic energy of the body.

NOTE

If the final E_k is smaller than the initial E_k of the object because it is slowing down, reverse the equation to produce a positive value for ΔE_k .

$$\text{i.e. } W = \Delta E_k = \frac{1}{2}mu^2 - \frac{1}{2}mv^2$$

Like all forms of energy, E_k is a scalar and has no direction associated with it.

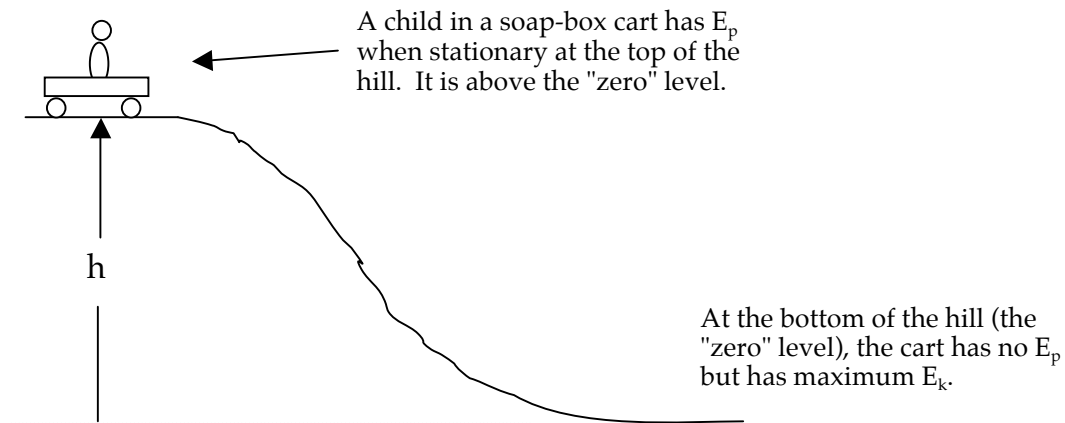
EXAMPLE

A car of mass $1.30 \times 10^3 \text{ kg}$ is moving along a road at 70.0 kmh^{-1} when it brakes sharply to avoid a person on a skateboard coming out of a driveway. The speed is reduced to 30.0 kmh^{-1} as the car swerves around the person. Calculate the work done by the brakes on the car.

GRAVITATIONAL POTENTIAL ENERGY

This is the energy possessed by a body due to its height above a given level.

Any object above a given "zero" level has the ability to release this energy as E_k as it falls to this "zero" level.



The amount of potential energy is given by:

$$E_p = mgh$$

where E_p = the potential energy of the body (J)
 m = the mass of the body (kg)
 g = the acceleration due to gravity (ms^{-2})
 h = the height above the "zero" level (m)

If an object is raised up to a height in the Earth's gravitational field, work must be done against the force of gravity.

This work is given by:

$$W = \Delta E_p = mg\Delta h$$

where Δh = the height through which the object is raised.

EXAMPLE

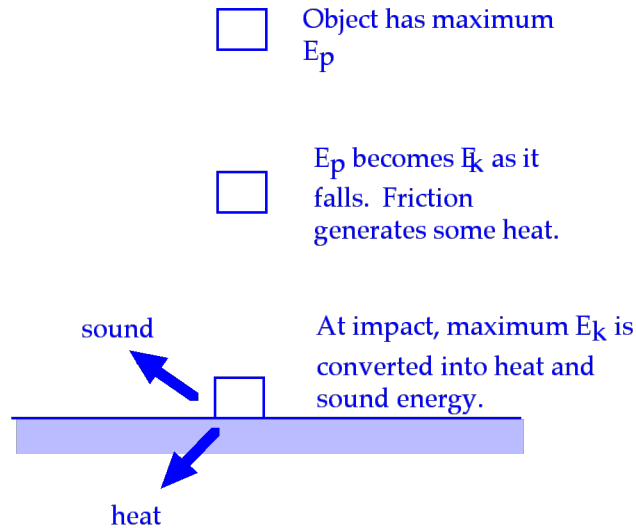
A cable car of combined mass 4.30×10^3 kg is used to transport 22 people from the carpark to the top of Grouse Mountain in Vancouver, a vertical rise of 457 m. Calculate the work done by the motors operating the car.

CONSERVATION OF MECHANICAL ENERGY

In all situations, energy must be conserved - it can be transformed from one form to another, but the total amount stays the same.

EXAMPLE

Consider an object dropped from a height onto a hard surface.



The above situation describes a particular case where the total mechanical energy is conserved.

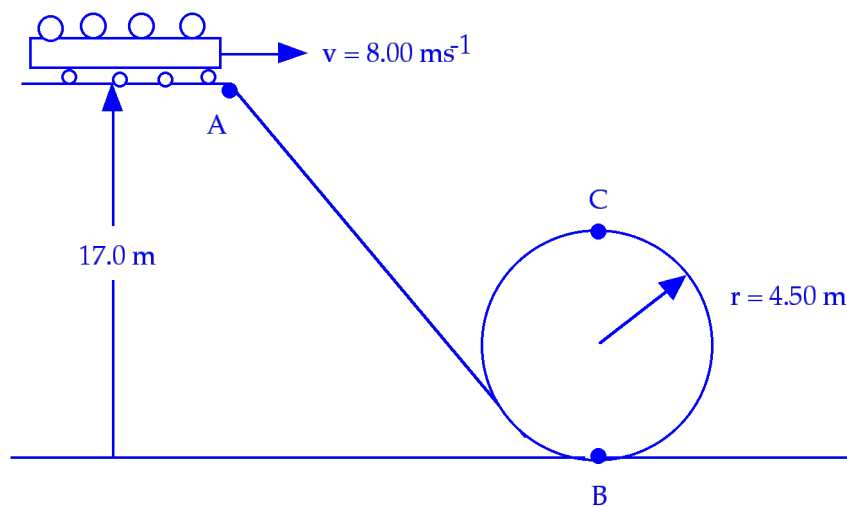
$$\begin{aligned} E_{\text{total}} &= E_p + E_k \\ &= mgh + \frac{1}{2}mv^2 \end{aligned}$$

NOTE

In calculations, it will generally be considered that friction is not acting.

EXAMPLE

A roller coaster of mass $3.70 \times 10^3 \text{ kg}$ is moving at 8.00 ms^{-1} at point A as shown below.



- (a) Determine the total energy of the coaster at A.
- (b) Calculate the speed achieved at point B.
- (c) Does the coaster safely make it through the loop if the minimum speed to do so is 6.64 ms^{-1} ? Justify your answer by calculation.

ELASTIC AND INELASTIC COLLISIONS

In all collisions, momentum is conserved. So too is energy (in all its different forms). However, it is unusual for a particular type of energy to be conserved during a collision.

ELASTIC COLLISION - one where no kinetic energy is lost, and the E_k and momentum of the system are both conserved.

e.g. In the interaction between atoms and sub-atomic particles.

Elastic collisions do not exist in everyday situations - only the sub-atomic.

INELASTIC COLLISION - one where momentum is conserved but not E_k (which is transformed into other forms of energy).

e.g. Collisions between billiard balls, cars, football players, etc.

EXAMPLE

A billiard ball of mass 0.150 kg moving at 1.20 ms^{-1} collides head-on with a similar ball moving at 0.900 ms^{-1} in the opposite direction. After the collision the first ball becomes stationary.

- (a) Calculate the final velocity of the second ball.
- (b) Compare the sum total of the initial and final kinetic energies.